Chapter 8: Magnetism

Textbook: Fundamentals of Physics by Halliday, Resnick & Walker [extended 9th edition]

Topics to be Covered:

*28-2: What Produces a Magnetic Field, 28-3: The Definition of **B**(Finding the Magnetic Force on a Particle, Magnetic Field Lines), 29-2: Calculating Magnetic Field Due to a Current (Magnetic Field Due to a Current in a Long Straight Wire), 29-4: Ampere's Law (Magnetic Field Outside a Long Straight Wire with Current).*

Theory:

1. Explain magnetic force, magnetic field and magnetic field lines.
2. State and explain Biot-Savart Law and Ampere's Law.
3. Find out the magnetic field due to a current in a long straight wire using Biot-Savart Law.
4. Derive the magnetic field outside a long straight wire by Ampere's Law.

Problems:

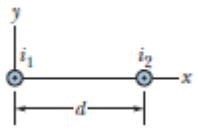
Chapter-28

1. [1] A proton traveling at 23.0° with respect to the direction of a magnetic field of strength 2.60 μT experiences a magnetic force of 6.50×10^{-17} N. Calculate (a) the proton's speed and (b) its kinetic energy in electron-volts.
2. [2]A particle of mass 10 g and charge 80 μC moves through a uniform magnetic field, in a region where the free-fall accelerationis $-9.8\hat{\jmath}$m/s^2 . The velocity of the particle is a constant$20\hat{\imath}$Km/s,which is perpendicular to the magnetic field. What, then, is themagnetic field?
3. [3] An electron that has velocity$\vec{v} = 2.0 \times 10^6\, ms^{-1}\hat{\imath} + 3.0 \times 10^6 ms^{-1}\hat{\jmath}$ moves through the uniform magnetic field$\vec{B} = 0.030\, T\, \hat{\imath} - 0.15\, T\, \hat{\jmath}$. Find the force on the electron due to the magnetic field.
4. [4] An alpha particle travels at a velocity of magnitude 550 m/sthrough a uniform magnetic field of magnitude 0.045 T. (An alphaparticle has a charge of $+3.2 \times 10^{-19}$ C and a mass of 6.6×10^{-27} kg.) The angle between ***v***and ***B***is 52°.What is the magnitudeof (a) the force $\boldsymbol{F_B}$ acting on the particle due to the field and (b) the acceleration of the particle due to $\boldsymbol{F_B}$?

Chapter-29

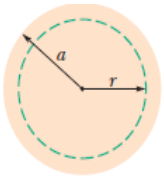
5. [3] At a certain location in the Philippines, Earth's magnetic field of 39 μT is horizontal and directed due north. Suppose the net field is zero exactly 8.0 cm above a long, straight, horizontal wire that carries a constant current. What is the magnitude of the current?
6. [9] Two long straight wires are parallel and 8.0 cm apart. They are to carry equal currents such that the magnetic field at a point halfway between them has magnitude 300 μT. How much current is needed?

7. [12] In Figure, two long straight wires at separation $d = 16.0\ cm$ carry currents $i_1 = 3.61$ mA and $i_2 = 3.00i_1$ out of the page. Where on the x axis is the net magnetic field equal to zero?



8. [42] In a particular region there is a uniform current density of 15 A/m^2 in the positive z direction. What is the value of $\oint \boldsymbol{B}.\boldsymbol{ds}$ when that line integral is calculated along the three straight-line segments from (x, y, z) coordinates (4d, 0,0) to (4d, 3d, 0) to (0, 0, 0) to (4d, 0, 0), where d = 20 cm?

9. [43] Figure shows a cross section across a diameter of a long cylindrical conductor of radius a = 2.00 cm carrying uniform current 170 A. What is the magnitude of the current's magnetic field at radial distance 4.00 cm?



10. [74] The magnitude of the magnetic field 88.0 cm from the axis of a long straight wire is 7.30 μT. What is the current in the wire?